YAŞAR UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING

COMP4910 Senior Design Project 1, Fall 2025
Advisor: Arman Savran

WQ: Wander Quest
Final Report

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PLAGIARISM STATEMENT

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KEYWORDS

Travel Motivation, Gamification, Geolocation, Reward System, Mobile Application, Interactive Map, Travel Visualization, AI Recommendations, Machine Learning, Unity 3D.

ABSTRACT

Wander Quest is a mobile application designed to enhance travel experiences by enabling users to log their journeys, visualize their travels on an interactive map, and engage in gamified activities. Wander Quest keeps users motivated with features like daily and long-term tasks, as well as sponsored missions. Users earn rewards such as discounts and can compete with friends for higher rankings. Using geolocation APIs and a database, Wander Quest provides an engaging platform that promotes exploration and community.

ÖZET

Wander Quest, kullanıcıların seyahat deneyimlerini geliştirmeyi hedefleyen oyunlaştırılmış bir mobil uygulamadır. Geolokasyon API'leri ve dinamik ödül sistemi entegre ederek, kullanıcıların ziyaret edilen yerleri kaydetmelerine, yeni bölgelerin kilidini açmalarına ve keşiflerle ödüller kazanmalarına olanak tanır. Wander Quest kullanıcıların seyahatleri için görev sistemi ile birlikte arkadaşlarıyla rekabet etmesine olanak sağlayarak, seyahat motivasyonu ele alarak etkileşimli ve ilgi çekici bir seyahat topluluğu oluşturmayı amaçlar.

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LIST OF FIGURES

No figures are used in this project documentation.

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LIST OF ACRONYMS/ABBREVIATIONS

UML	Unified Modeling Language
API	Application Programming Interface
AI	Artificial Intelligence
PAF	Project Assignment Form
RSD	Requirements Specification Document
DSD	Design Specification Document

1. INTRODUCTION

1.1. Description of the Problem

Modern travellers often forget their travel history, and many lack the motivation to explore new places or stay active. Wander Quest addresses this issue by providing a platform that combines travel logging with gamification to enhance user motivation and engagement.

Section 2 provides an survey of related work, while Section 3 and Appendix A outline the problem definition and requirements in detail. The high-level design and corresponding design decisions are discussed in Section 4 and Appendix B. Project management aspects are addressed in Section 5 and Appendix C, and lastly, Section 6 concludes the report and highlights potential future work.

1.2. Project Goal(s)

A gamified mobile application called Wander Quest was developed to encourage users to explore new locations using geolocation APIs and a dynamic reward system. The project is divided among the teammates balanced and according to the topic they are interested in and be good at. Unity 3D was used by game developer teammates to bring this concept to life. Another team member worked on creating an efficient user tracking system and database to store travel history, along with implementing AI-based recommendations.

1.3. Project Outputs/Deliverables

Project outputs for COMP 4910 project:
Documentation:

- Project Assignment Form (PAF)
- Requirements Specification Document (RSD) v1.0.
- Requirements Specification Document (RSD) v2.0.
- Design Specification Document (DSD) v1.0.
- Project poster
- Project presentation
- Web site

Application:

- Integration with geolocation APIs for accurate tracking
- A mobile application featuring travel logging and its visualization.
- Gamification elements, including a quest system with daily, side, and main quests.
- A dynamic user interface that incorporates 3D interactive maps for enhanced engagement.
- Integration of Firebase as the backend architecture to store and analyse user data efficiently.

Predicted additional outputs for COMP 4920:
Documentation:

- Requirements Specification Document (RSD) v3.0.

Application:

- Fully completed Wander Quest mobile application.
- Personalized recommendations and suggestions for future explorations using AI.
- Support for multilingual localization to cater to a global audience.
- Leaderboard and friends system implementation.
- World map feature and colorized visited locations implementation.

2. SURVEY OF RELATED WORK

Existing travel logging and gamification-based applications provide insights into how organizations have approached the intersection of geolocation, gamification, and user engagement. While current solutions excel in specific domains, gaps remain in addressing the combined needs of travel tracking and motivation through interactive features.

Open System Products:

1. **OpenStreetMap**
   OpenStreetMap is a community-driven platform offering free geospatial data for travel logging and visualization. Its open API supports custom integrations, enabling developers to create tailored mapping solutions. While it provides extensive mapping capabilities, OpenStreetMap lacks features specifically designed for gamification or interactive user experiences. [2]

Commercial Products:

1. **Google Maps**
   Google Maps provides robust geolocation and map visualization capabilities, including route planning, real-time traffic updates, and location tagging. It is widely used for navigation and location-based services. However, its primary focus is on delivering geospatial data and navigation tools, rather than fostering user engagement or incorporating gamification features. [1]
2. **Snapchat Map**
   Snapchat's Map feature integrates geolocation with social interactivity, allowing users to share locations and view activity hotspots. This social dimension enhances engagement, but the feature is not designed to encourage exploration or reward users for visiting new places. [3]
3. **Pokémon GO**
   Pokémon GO blends augmented reality and geolocation to create a highly engaging mobile gaming experience. Players are motivated to explore real-world locations to capture Pokémon and complete challenges. While this approach successfully incentivizes outdoor activity, its primary focus is on gaming rather than travel logging or visualization. [4]

The reviewed solutions highlight a gap in applications that effectively combine travel logging with gamification to motivate exploration. Wander Quest bridges this gap by integrating travel tracking with an engaging quest system, social features like leaderboards, and personalized AI-driven recommendations. Unlike Pokémon GO, which focuses solely on gaming, or Google Maps, which prioritizes navigation, Wander Quest provides a rewarding platform that promotes exploration and travel logging through gamification.

3. REQUIREMENTS

The requirements for Wander Quest were initially determined during the Fall semester as part of COMP 4910. They were first outlined in a PAF, which included the project code, title, team details, and a summary of the project objectives.

Subsequently, RSD v1.0 was prepared to document the initial requirements in a structured textual format. These requirements were later refined and formalized in RSD v2.0, using UML notation to enhance clarity and organization.
The finalized requirements for COMP 4910, included in Appendix A: RSD v2.0, serve as the basis for the project's design and implementation.

In the spring semester for COMP 4920, it is expected that the requirements may be revised due to:

1. New Ideas: Adding features or improvements based on input from the project advisor and team.
2. Feasibility Adjustments: Removing requirements that are too complex to implement within the available time and resources.

Briefly planned requirements:

Core Features:

- Travel Logging: Log visited locations with geolocation integration.
- Travel Visualization: Interactive map with heatmap and timeline features.
- Area Unlocking System: Unlock new regions based on milestones.

Gamification:

- Tasks System: Encourage exploration through incentives.
- Sponsored Missions: Reward users for achievements.
- Leaderboards: Foster competition among users.

Database Features:

- Centralized Data Storage: Store user profiles, travel logs, and achievements securely in a Firebase database.
- Secure Access Control: Protect sensitive user information with authentication and authorization protocols.

Future Enhancements:

- AI-based personalized recommendations.
- Social features for community engagement.

4. DESIGN

4.1. High Level Design

The Wander Quest system employs a layered architecture to ensure modularity, scalability, and ease of maintenance according to the requirements of the project. The system is divided into three primary layers:

- **Presentation Layer**: Manages user-facing components, including the 3D map interface and quest interactions. This layer is implemented using Unity to deliver an engaging user experience.

- **Logic Layer**: Handles core functionalities such as quest management, map navigation, and geolocation services. This layer bridges the presentation and data layers to maintain seamless functionality.
- **Data Access Layer**: Interfaces with external APIs and databases, ensuring efficient storage and retrieval of data. This includes geolocation data from Google Maps and user information stored in Firebase.

The high-level design specifications are detailed in Appendix B: Design Specifications Document, where system architecture, component interactions, and UML diagrams are provided.

4.2. Detailed Design

The detailed design phase, planned for COMP 4920, will build on the high-level design by refining subsystems and adding advanced features. It will include all project requirements in detail. Since this is COMP 4910, the detailed design will be provided in the spring semester.

4.3. Realistic Restrictions and Conditions in the Design

The design of Wander Quest has several realistic restrictions and conditions to ensure feasibility and alignment with available resources and project goals:

- **Dependency on Stable Geolocation Services:** The functionality of the application relies heavily on accurate and stable geolocation services. Disruptions in these services could affect the user's experience.
- **API Resource Management:** The application uses geolocation APIs with free-tier limitations, permitting only 200$ usage. To support larger-scale usage, sponsorships or alternative funding sources will be required to cover costs. [6]
- **Data Storage Constraints:** The application utilizes Firebase Realtime Database's free tier, which has a total storage limit of 1GB, and a download is limited to 10GB per month. For further needs sponsorships or alternative funding sources will be required to cover costs. [5]
- **Sponsorship Requirements:** Collaboration with sponsors is necessary to fund in-app missions and cover ongoing project expenses, including API usage and maintenance.

These restrictions and conditions are reflective of the realistic constraints associated with resource availability, technological limitations, and the initial scope of the project.

5. IMPLEMENTATION AND TESTS

5.1. Implementation of the Product

Because the project is a software development project, software programs were used to help develop the application. The implementations completed in COMP 4910 will continue for the development in COMP 4920. The project will further utilize the following technologies:

- **Unity 3D:** For developing the entire application, including its gamification features.
- **Google Maps API:** To provide geolocation functionality and enhance user interaction with map-based features.
- **Firebase:** As the backend for efficient data storage, retrieval, and analysis.

The technologies were selected to best meet the project's objectives. Unity 3D was chosen for its strong tools and adaptability as a game engine, making it ideal for developing an interactive gamified application. Firebase was selected for its dependable and scalable backend, which is effective for managing user data. The Google Maps API was included for its advanced geolocation features and the valuable support it provides to developers.

5.2. Tests and Results of Tests

In this project, testing is critical to ensure a secure, reliable, and user-friendly application for the users. Based on the application's requirements and features, these tests are determined as necessary:

Functionality tests
- Geolocation accuracy
- Mission systems including daily, side, and main quests
- Leaderboard tracking
- Friend-adding system
- Map colorization

Usability tests
- Interface navigation
- Map navigation

Performance tests
- Database interactions
- API interactions

Security tests
- User data handling
- Login system implementation

Additional tests
- Multilingual localization
- AI-based personalized recommendations

6. PROJECT MANAGEMENT

6.1. Project Plan

The project plan for the Fall semester of COMP 4910 includes the following key activities and their respective schedules. The tasks are structured as follows:

Task	Start Week	End Week	Duration (Weeks)
Setup and Research	Week 1	Week 6	6
Map Visualization	Week 4	Week 10	5
Quest System	Week 10	Week 12	3
Login System	Week 11	Week 13	3
Map Features	Week 9	Week 14	4
UI Design	Week 10	Week 13	4
Deliverables	Week 7	Week 15	3

This plan is detailed in Appendix C.

6.2. Project Effort/Manpower

Team Member	Total Hours Spent	Total Man-Days
Ömer Ünal	190	23.75
Tahsin Berk Öztekin	190	23.75
Seden Canpolat	190	23.75
Team Total	570	71.25

Effort is detailed for each month in Appendix C, showing individual contributions per month and consolidated team totals.

6.3. Project Cost Analysis

The project primarily relies on software and APIs, resulting in minimal hardware expenses. Below is a breakdown of the project costs:

- **Google Maps API Access**: The Google Maps API provides $200 of free usage each month as part of its free tier. For services like Maps, Routes, and Places APIs, costs can escalate significantly depending on the number of requests and active users. For instance, Places API costs $32 per 1,000 requests, and Routes API starts at $5 per 1,000 requests. Current project usage remains within the free tier; however, future costs could increase substantially with more users. [6]
- **Development Tools**:
  - Unity: Free for small projects.
  - Visual Studio Code: Free.
- **Database Hosting**: Firebase offers a free Spark Plan that includes 1 GB of storage and 50,000 document reads per day. For future scalability, the Blaze Pay-As-You-Go Plan is estimated at $50/month, depending on storage and read/write operations. This plan allows greater flexibility for larger-scale applications. [5]

Manpower effort was not monetized, as the contributions were made by team members as part of the course requirements. Other costs such as travel, communication, or office expenses were negligible or not applicable in this project.

7. CONCLUSIONS

7.1. Summary

Wander Quest is a gamified mobile application designed to motivate users to explore new locations by integrating travel logging, geolocation APIs, and interactive quests. Throughout the development process, the project team has completed research on existing solutions, identified key requirements, and designed an engaging user experience. The implementation phase included the integration of Firebase for secure data storage, implementation and usage of Google Maps API for maps features and Unity 3D for a dynamic and interactive interface. Initial testing has been conducted to evaluate core functionalities such as geolocation tracking, quest mechanics and login system. Moving forward, further refinements and additional features will be developed in COMP 4920 to enhance the application's scalability, performance, and user engagement.

7.2. Benefits of the Project

Wander Quest is an application that provides numerous benefits to its users, to society, to the environment and to the economy.
Numerous benefits of Wander Quest in different areas:
- **Users**: Encourages exploration, promotes physical activity, and enhances travel experiences through gamification.
- **Society**: Fosters social interaction and discovery of new cultural and historical locations.
- **Environment**: Encourages sustainable tourism by promoting local travel and less-visited destinations.
- **Economy**: Offers monetization opportunities through partnerships and sponsored quests.

7.3. Future Work

Future work can be split into the two parts as planned ones for COMP 4920 and the ones in the distant feature:

Planned for COMP 4920:
- **AI Support:** Enhancing AI-driven recommendations, and refining quest mechanics.
- **Multilingual support:** Adding multilingual support for users all around the world.
- **Social Features**: Implementing a friend system and expanding leaderboard functionalities.
- **Sponsorship Features**: Reward users for achievements with sponsored quests and at the same time make the advertisement of sponsors.

Potential Future Enhancements:
- **Wearable Device Support**: Enabling compatibility with smartwatches.
- **iOS Version**: Expanding to iOS for broader accessibility.

REFERENCES

[1] https://www.google.com/maps
[2] https://www.openstreetmap.org/#map=6/39.03/35.25
[3] https://www.snapchat.com/
[4] https://pokemongolive.com/
[5] https://firebase.google.com/pricing?hl=en&authuser=0
[6] https://mapsplatform.google.com/pricing/
[7] https://developers.google.com/maps/documentation/places/web-service?hl=tr
[8] https://docs.unity.com/
[9] https://firebase.google.com/docs?hl=tr
[10] https://developers.google.com/maps/documentation?hl=tr

APPENDICES

APPENDIX A: REQUIREMENTS SPECIFICATION DOCUMENT

Assume 10 pages of RSD
will be placed here
as Pages 8-17

APPENDIX B: DESIGN SPECIFICATION DOCUMENT

Assume 10 pages of DSD
will be placed here
as Pages 19-28

APPENDIX C: PROJECT MANAGEMENT DOCUMENTS

APPENDIX C1: PROJECT PLAN

COMP 4910/4920 WQ: Wander Quest, Project Plan, 18.01.2025, v1.1																	
Task No	**Task Name**	**Weeks**															**Any Additional Notes**
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	
1	Setup and Re-search	X	X	X	X	X	X			X							
2	Map Visualiza-tion				X	X	X			X	X						
3	Quest System										X	X	X				
4	Login System											X	X	X			
5	Map Features									X	X		X		X		
6	UI Design										X	X	X	X			
7	Deliverables							X							X	X	
8																	

APPENDIX C2: PROJECT EFFORT LOG- CONSOLIDATED

COMP 4910 Project Effort Log, Project Code: WQ, 18.01.2025, v1.0								
Week	**Dates**	**Ömer Ünal**		**Tahsin Berk Öztekin**		**Seden Canpolat**		**Total Weekly Effort in Man-Hours**
		Work Done	**Total Hours Spent**	**Work Done**	**Total Hours Spent**	**Work Done**	**Total Hours Spent**	
Week 1	6/10/24	Project planning.	16	Project planning.	16	Project planning.	16	48
Week 2	13/10/24	Idea presentation.	6	Idea presentation.	6	Idea presentation.	6	18
Week 3	20/10/24	Requirement analysis.	18	Requirement analysis.	18	Requirement analysis.	18	54
Week 4	27/10/24	Project planning	16	Project planning.	16	Project planning.	16	48
Week 5	3/11/24	Base setup; map visualization.	16	Project planning.	16	Project planning.	16	48
Week 6	10/11/24	Route exploration; Unity setup.	14	Firebase research and setup.	14	Explored Google Maps and Places APIs.	14	42
Week 7	17/11/24	RSD form creation.	10	RSD form creation.	10	RSD form creation.	10	30
Week 8	24/11/24	-	0	-	0	-	0	0
Week 9	1/12/24	Places API; Tiles API integration.	18	Firebase setup; data structure design.	18	Processed gyroscope data; added orientation arrow.	18	54
Week 10	8/12/24	Map movement; quest system.	15	Quest data integration with Firebase.	15	Implemented quest UI; created notifications.	15	45
Week 11	15/12/24	Map-UI merge; quest features; tracking tests.	16	User-specific database; login system.	16	Completed quest journal UI; added touch navigation.	16	48
Week 12	22/12/24	Login UI development.	16	User authentication.	16	Fixed gyroscope bugs; improved sensors.	16	48
Week 13	29/12/24	Bug fixes; Google Sign-In.	16	Firebase fixes; Google Sign-In.	16	Worked on directional arrow.	16	48
Week 14	5/1/25	Project deliverables.	13	Project deliverables.	13	Project deliverables.	13	39
Total Effort in Man-Hours			**190**		**190**		**190**	**570**
Total Effort in Man-Days			**23.75**		**23.75**		**23.75**	**71.25**

APPENDIX C3: PROJECT EFFORT LOGS FOR EACH TEAM MEMBER-

Assume 3 pages for 3 Team Members
Pages 34, 35, 36

COMP 4910/4920 Project Effort Log, Project Code: WQ, 18.1.2025, v1.0
Team Member: Ömer Ünal

Week	Dates	Work Done in Some Detail	Total Hours Spent
Week 1	6/10/24	Planned the project	16
Week 2	13/10/24	Presented our project idea to our advisor	6
Week 3	20/10/24	Requirement analysis was made	18
Week 4	27/10/24	Planned the details of the project	16
Week 5	3/11/24	Started the base of the project. Worked on map visualization.	16
Week 6	10/11/24	Investigated route visualization on the map; set up Unity and GitHub.	14
Week 7	17/11/24	Contributed to creating the RSD form.	10
Week 8	24/11/24	Midterm Week	0
Week 9	1/12/24	Implemented Places API for displaying nearby locations as 3D points; started using Tiles API.	18
Week 10	8/12/24	Implemented map movement, zoom, and pin placement; created the basic quest system.	15
Week 11	15/12/24	Merged map and main UI; added functionality for quest creation, searching, and tracking; tested location tracking.	16
Week 12	22/12/24	Created Login UI frontend.	16
Week 13	29/12/24	Resolved login and build bugs; worked on Google Sign-In integration.	16
Week 14	5/1/25	Worked on the project deliverables	13
Total Effort in Man-Hours			**190**
Total Effort in Man-Days			**23.75**

Notes:

1. This table shows the team member project effort. One Man-Day is Eight Man-Hours.
2. Each team member must fill out the form periodically (preferably at the end of the week of any work done).
3. Each filled-out table must be emailed at the end of each month to a selected Project Member (cc to Project Advisor), who will produce a consolidated table.

COMP 4910/4920 Project Effort Log, Project Code: WQ, 18.1.2025, v1.0
Team Member: Tahsin Berk Öztekin

Week	Dates	Work Done in Some Detail	Total Hours Spent
Week 1	6/10/24	Planned the project	16
Week 2	13/10/24	Presented our project idea to our advisor	6
Week 3	20/10/24	Requirement analysis was made	18
Week 4	27/10/24	Planned the details of the project	16
Week 5	3/11/24	Planned the details of the project	16
Week 6	10/11/24	Researched and began setting up Firebase for data management.	14
Week 7	17/11/24	Contributed to creating the RSD form.	10
Week 8	24/11/24	Midterm Week	0
Week 9	1/12/24	Completed Firebase database setup; analysed data structures; designed efficient models for data storage.	18
Week 10	8/12/24	Integrated quest data creation and synchronization with Firebase.	15
Week 11	15/12/24	Designed user-specific database branches; implemented a simplified login system.	16
Week 12	22/12/24	Enhanced user authentication; added quest functionalities through Firebase.	16
Week 13	29/12/24	Resolved asynchronous Firebase sync issues; began integrating Google Sign-In.	16
Week 14	5/1/25	Worked on the project deliverables	13
Total Effort in Man-Hours			**190**
Total Effort in Man-Days			**23.75**

Notes:

1. This table shows the team member project effort. One Man-Day is Eight Man-Hours.
2. Each team member must fill out the form periodically (preferably at the end of the week of any work done).
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COMP 4910/4920 Project Effort Log, Project Code: WQ, 18.1.2025, v1.0 Team Member: Seden Canpolat			
Week	**Dates**	**Work Done in Some Detail**	**Total Hours Spent**
Week 1	6/10/24	Planned the project	16
Week 2	13/10/24	Presented our project idea to our advisor	6
Week 3	20/10/24	Requirement analysis was made	18
Week 4	27/10/24	Planned the details of the project	16
Week 5	3/11/24	Planned the details of the project	16
Week 6	10/11/24	Explored Google Maps API and Places API for project requirements.	14
Week 7	17/11/24	Contributed to creating the RSD form.	10
Week 8	24/11/24	Midterm Week	0
Week 9	1/12/24	Processed gyroscope data for user orientation; added an on-screen arrow to guide users.	18
Week 10	8/12/24	Implemented quest UI functionality; created functional notifications.	15
Week 11	15/12/24	Completed quest journal UI; implemented touch gesture-based map navigation.	16
Week 12	22/12/24	Fixed gyroscope-related bugs, improving sensor reliability.	16
Week 13	29/12/24	Began working on a directional arrow for targeted locations.	16
Week 14	5/1/25	Worked on the project deliverables	13
Total Effort in Man-Hours			**190**
Total Effort in Man-Days			**23.75**
Notes:			
1. This table shows the team member project effort. One Man-Day is Eight Man-Hours. 2. Each team member must fill out the form periodically (preferably at the end of the week of any work done). 3. Each filled-out table must be emailed at the end of each month to a selected Project Member (cc to Project Advisor), who will produce a consolidated table.			